Calculus II Assignment 6

Part I. Regular Problems

1. Express each linear function as $f(\mathbf{X}) = \mathbf{C} \cdot \mathbf{X}$ for some vector $\mathbf{C}$ or $\mathbf{F}(\mathbf{X}) = \mathbf{C}\mathbf{X}$ for some matrix $\mathbf{C}$.

   (a) $f(x_1, x_2) = x_1 + 2x_2$

   (b) $f(x_1, x_2, x_3) = x_1 + 2x_2$

   (c) $\mathbf{F}(x_1, x_2) = (x_1, x_1 + 2x_2)$

   (d) $\mathbf{F}(x_1, x_2, x_3) = (x_1, x_1 + 2x_2)$

   (e) $\mathbf{F}(x_1, x_2, x_3, x_4) = (x_4 - x_2, x_3 - x_1, x_2 + 5x_1, x_1 + x_2 + x_3 + x_4)$

   (f) $\mathbf{F}(x_1, x_2) = (x_1, 5x_1, -x_2, -2x_1, x_2)$

   (g) $\mathbf{F}(x_1, x_2, x_3, x_4, x_5) = (x_1, 5x_1, -x_2, -2x_1, x_2)$

2. The function

$$f(x, y) = 5 + x + x^2 + y^2$$

   is the sum of the constant function 5 , the linear function x, and the degree 2 polynomial $x^2 + y^2$.

   (a) Give two examples of points (x, y) near $(0, 0)$ for which the approximation by the first two parts,

$$f(x, y) \approx 5 + x$$

   differs from $f(x, y)$ less than $\frac{1}{100}$.

   (b) Show that

$$g(u, v) = f(1 + u, 2 + v)$$

   is the sum of a constant function, a linear function of (u, v), and a degree 2 polynomial function of u and v.

3. Let $f(\mathbf{X}) = \|\mathbf{X}\|^2, \mathbf{X}$ in $\mathbb{R}^4$. Let $\mathbf{A}$ be in $\mathbb{R}^4$ and define

$$g(\mathbf{X}) = \|\mathbf{A}\|^2 + 2\mathbf{A} \cdot (\mathbf{X} - \mathbf{A})$$

   (a) Let $\mathbf{U} = \mathbf{X} - \mathbf{A}$. Use the formula

$$\|\mathbf{A} + \mathbf{U}\|^2 = \|\mathbf{A}\|^2 + 2\mathbf{A} \cdot \mathbf{U} + \|\mathbf{U}\|^2$$

   to show that the difference between $f(\mathbf{X})$ and $g(\mathbf{X})$ is $\|\mathbf{U}\|^2$.

(b) Show that the difference between $f(\mathbf{X})$ and $g(\mathbf{X})$ does not exceed 10^{-4} when $\|\mathbf{X} - \mathbf{A}\| < 10^{-2}$

4. Sketch or describe the c level sets, $f(\mathbf{X}) = c$ in $\mathbb{R}^3$, for the following functions f from $\mathbb{R}^3$ to $\mathbb{R}$ and values c.

   (a) $f(\mathbf{X}) = \|\mathbf{X}\|^2, c = \frac{1}{2}, 1, 2$

   (b) $f(\mathbf{X}) = \|\mathbf{X}\|, c = \frac{1}{2}, 1, 2$

   (c) $f(\mathbf{X}) = \frac{1}{\|\mathbf{X}\|^2}, \quad c = \frac{1}{2}, 1, 2$

5. The slope of the graph of $\cos(cx)$ lies in the interval $[-c, c]$. Fill in the missing numbers.

   (a) if $|x - a| < (?)$ then $|\cos(2x) - \cos(2a)| < \epsilon$.

   (b) if $|y - b| < \delta$ then $|\cos(3y) - \cos(3b)| < (?)$.

   (c) if $\|(x, y) - (a, b)\| < (?)$ then $|\cos(2x)\cos(3y) - \cos(2a)\cos(3b)| < \epsilon$.

6. Show that the function
$$f(x, y, z) = \frac{\sin(x^2 + y^2)}{e^{z+y}}$$
is continuous at all points (x, y, z).

Part II. Group Problems

1. Sketch or describe the c level sets, $f(x, y) = c$ in $\mathbb{R}^2$, for the following functions f and values c. Use the level sets to help sketch the graph of the function.

   (a) $f(x, y) = x + 2y, c = -1, 0, 1, 2$

   (b) $f(x, y) = xy, c = -1, 0, 1, 2$

   (c) $f(x, y) = x^2 - y, c = -1, 0, 1, 2$

2. Determine the set of points at which the function is continuous.

   (a)
$$f(x, y) = \frac{xy}{1 + e^{x-y}}$$

   (b)
$$f(x, y) = \frac{1 + x^2 + y^2}{1 - x^2 - y^2}$$

   (c)
$$f(x, y, z) = \arcsin\left(x^2 + y^2 + z^2\right)$$